

CSE-214 Online

Behavioural Design Pattern

Duration: 30 Minutes

Problem: Smart Discount Calculation System

An e-commerce platform wants to provide customers with the best possible discount during checkout.

For every purchase, the system considers three different discount policies:

1. Purchase Amount Discount
2. Customer Category Discount
3. Payment Method Discount

Each policy calculates a discount percentage independently. However, the discounts are not added together.

After evaluating all applicable policies, the system applies **only the highest discount percentage** to the purchase.

1. Purchase Amount Discount

The discount depends on the total purchase amount.

For every complete ₪1,000 of purchase, the customer receives an additional **5% discount**, up to a maximum discount of **25%**.

Purchase Amount	Discount
Below ₪1,000	0%
₪1,000 - ₪1,999	5%
₪2,000 - ₪2,999	10%
₪3,000 - ₪3,999	15%
₪4,000 - ₪4,999	20%
₪5,000 or more	25%

For example:

Purchase Amount = ₪3,600

Purchase Amount Discount = 15%

2. Customer Category Discount

Customers belong to one of the following categories:

REGULAR -> 5%

PREMIUM -> 15%

For example, a Premium customer is eligible for a **15% discount** regardless of the purchase amount.

3. Payment Method Discount

The platform provides different discounts depending on the selected payment method.

CARD -> 2%

MFS -> 5%

CASH -> 8%

MFS includes mobile financial services such as **bKash** and **Rocket**.

Final Discount Selection

For every purchase, all applicable discount policies must be evaluated independently. If two or more policies provide the same maximum discount, any one of those policies may be considered the selected policy since the final payable amount will be the same.

The system then selects the **highest discount percentage** among them.

The discounts must not be combined.

For example, consider the following purchase:

Purchase Amount : ₪3,500
Customer Type : PREMIUM
Payment Method : CASH

The available discounts are:

Purchase Amount Discount -> 15%
Customer Category Discount -> 15%
Payment Method Discount -> 8%

Therefore:

Applied Discount = 15%

The final payable amount is calculated using only this discount.

Another Example

Consider:

Purchase Amount : ₪5,500
Customer Type : PREMIUM
Payment Method : CASH

The discount calculations are:

Purchase Amount Discount -> 25%
Customer Category Discount -> 15%
Payment Method Discount -> 8%

Therefore:

Applied Discount = 25%

Only the **25% purchase amount discount** is applied.